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Analysis of collective dynamics of particulate systems modeled by Markov chains

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ABSTRACT

This paper develops an approach to capturing and analyzing the collective dynamics of particulate systems, exemplified using a horizontal rotating drum. The dynamics of a particulate system is modeled stochastically via the Markov chains approach using data obtained from Discrete Element Method simulations. Quantitative measures are developed to characterize the features of collective dynamics based on eigenvalues and singular values analysis of the Markov chain operators. These features include the dynamic modes corresponding to the oscillatory behavior and the spatial distribution of particle movements. It is also shown that these quantitative measures can be linked to the classification of flow regimes in a horizontal rotating drum.

1. Introduction

The behaviors of particulate systems are complex which can be studied at different scales, ranging from the microscopic to the macroscopic scale. Microscopic studies deal with the dynamic description of individual particle within the system. The Discrete Element Method (DEM), by solving Newton's laws of motion for each particle, has been used in many microscopic studies to obtain detailed information of particle behaviors including particle location, velocity and forces acting on each individual particle at any time during the process [1–6]. The results from DEM simulations focus on the microscopic scale and no measure of macroscopic behavior from DEM has been developed. However, in practice, it is often the macroscopic scale or the collective (overall) dynamics of the process that is important in particulate operations, e.g., particulate mixing, segregation and avalanche [7,8].

Macroscopic studies of particulate systems, such as particulate mixing and flow regime characterization, have been reported in the literature. Horizontal rotating drums are one of the most common applications considered in the literature as it is used in pharmaceutical, mining and chemical industries. The work that links the bed shape behavior of the particle motion in a horizontal rotating drum to the Froude number (which is the ratio of the centrifugal force to gravity) based on experimental studies is presented in [9]. The experimental analysis of rolling and slumping flow regimes in relation to the angle of repose for a horizontal rotating drum system is studied in [10]. The experimental study of particulate flow regimes in a horizontal rotating drum based on the relationship between Froud number, filling degree and wall-friction coefficient is presented in [11]. The study of particulate behaviors as functions of drum rotational speed, fill level and particulate size using a digital recording device can be found in [12]. Solidification technique, a method which uses chemicals to solidify particulate system in order to maintain the particle coordination, coupled with computerized image analysis, is used for experimental study of flow regimes and particulate mixing in a horizontal rotating drum [13]. While the approaches mentioned above provide useful qualitative and coarse quantitative analysis of steady-state overall particle behavior, they are not capable in capturing dynamical features of the collective particle motions.

There are some studies on using Markov chains, a type of stochastic model, to describe the overall dynamics of particulate systems. Markov chain models were developed based on experimental data to estimate particle distributions in a motionless (static) mixer [14] and to study particulate mixing in a hoop mixer [15]. The Markov chains approach was also used to model grinding processes [16]. A comprehensive review of these approaches can be found in [17]. The Markov chain operator can also be constructed from the microscopic information of particulate behavior generated by DEM simulations [18]. There are only few studies on analysis of Markov chain models reported in literature. The eigenvalues of the Markov chain operator were used to study the rate of convergence of the overall system behavior, e.g., the time needed for the system to reach a stationary particle distribution [19,20].

This work attempts to provide a systematic approach to determine more detailed *dynamic* features of the collective behavior (macroscopic) of particulate systems from the corresponding Markov chain operator. The eigenvalues analysis on Markov chain operators [18–20] is extended to obtain the magnitudes and frequencies of the key dynamic modes

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(components of the overall dynamics) and an index of overall system oscillatory behaviors. Singular Value Decomposition (SVD) is also used to determine the spatial distributions of particle movements in a horizontal rotating drum for various rotational speeds. In this work, Markov chain operators are constructed from data obtained from DEM simulations as in [18]. The results from the proposed approaches are compared with the qualitative analysis of flow regimes reported in [1,21,22].

The paper is organized as follows. In Section 2, the DEM, developed from Newton's laws of motion, for horizontal rotating drums is reviewed. A review of Markov chains theory used in this work is presented in Section 3. Analytical techniques to extract collective dynamical features from Markov chain operators including the illustrative example are presented in Section 4. Discussion of the proposed approaches is presented in Section 5. Conclusions and future work are discussed in Section 6.

2. DEM simulation and flow regimes

This section describes the DEM simulations which are used to generate particle trajectory data for the construction of Markov chain operators. The classes of qualitative flow regimes for particulate systems in a horizontal rotating drum are also presented.

2.1. Data generation

Particulate systems typically involve a large number of particles, where each particle behaviors such as translational and rotational motions, can be studied based on Newton's second law of motion. Due to the interactions between particles and also interactions with the system boundary, such as the drum wall in a horizontal rotating drum mixer, the collective dynamics behavior of the particulate system is highly complex. DEM simulations are used to generate each particle trajectory by solving the differential equations obtained from Newton's law of motion [1–6].

For the system under study, Newton's second law of motion is used to describe the translational and rotational motion at each time t of a particle of radius R_i and mass m_i, leading to a system of differential equations given by

$$m_i \frac{d\mathbf{v}_i}{dt} = \sum_{i,j=0}^{n} \left(\mathbf{F}_{ij}^{n} + \mathbf{F}_{ij}^{s} + m_i g \right) \tag{1}$$

$$I_i \frac{d\theta_i}{dt} = \sum_{i,j=0}^{n} \mathbf{R}_i \times \mathbf{F}_{ij}^{s} - \mu_r R_i \left| \mathbf{F}_{ij}^{n} \right| \bar{\theta}_i, \tag{2}$$

where $\mathbf{v}_i$, θ_i and I_i are the transitional and angular velocities, and moment of inertia of particle i, respectively, while $\bar{\theta}_i$ represents a unit vector equal to θ_i divided by its magnitude. The vector $\mathbf{R}_i$ is running from the center of the particle to the contact point with its magnitude equal to the particle radius R_i. The first part of the right hand side in Eq. (2) represents the torque due to tangential force $\mathbf{F}_{ij}^{s}$, and the second part represents the rolling friction torque arising from the elastic hysteresis loss or viscous dissipation, where μ_r is the coefficient of rolling friction. The quantities $\mathbf{F}_{ij}^{n}$ and $\mathbf{F}_{ij}^{s}$ represent, respectively, the normal contact force and the tangential contact force imposed on particle i by particle j and are used to quantify particle collisions. Detailed description of the forces can be found in [21,23].

Eqs. (1) and (2) for all particles are solved simultaneously using three dimensional DEM, taking into account the interactions between each particles and the boundary, and hence provide detailed microscopic information of the system, including individual particle tracking, velocity and forces acting on each particles at any time during the process. The DEM has been used in various studies of particulate systems including in horizontal rotating drum applications. The results obtained from the DEM model for horizontal rotating drums have been validated in literatures and the DEM model used in the present work follows the one proposed in [1,21,23].

The DEM is used in this work to generate microscopic data of particulate systems to construct the Markov chains system from which the macroscopic particulate system information can be extracted. The simulation conditions used in this work are given in Table 1.

2.2. Flow regimes

Flow regimes are one of the commonly used qualitative collective features of particulate behaviors in horizontal rotating drums. Several types of flow regimes which are often encountered in horizontal rotating drum applications are rolling, cascading, cataracting and centrifuging regimes [9], as illustrated in Fig. 1. The drum rotational speed, particle-drum size ratio and particle properties (size, density, etc.) are the main factors that affect the flow regimes [21]. In this paper, we illustrate the proposed analysis by studying the effect of drum rotational speed.

Many industrial particulate processes are required to operate under a specific flow regime to meet product specification [11]. For example, grinding processes are best achieved at cataracting regime since it has the highest particulate impacts. While centrifuging regime is avoided for grinding processes as it leads to poor particulate impacts due to low particle interactions [22,24].

In general, flow regimes can be related to the collective dynamics of oscillatory behaviors of particle distribution and are qualitatively classified as such. In the present study, qualitative analysis of flow regimes is used as a benchmark to evaluate the quantitative approach proposed.

The next section presents the Markov chains theory and the construction of Markov chain operator based on data obtained from DEM simulations.

3. Particulate system models using Markov chains

This section briefly reviews the Markov chain theory and presents the construction of Markov chain operators based on data obtained from DEM simulation used in [1,21,23] following the approach proposed in [18].

3.1. Markov chains

In this work, trajectories of the particles in a rotating drum are modeled stochastically. The location of each individual particle at time t varies for consecutive repetition of the same drum operating conditions. Detailed information regarding stochastic systems can be found in [25,26]. Markov processes are the simplest type of stochastic processes [27], where only the present states (but not past states) affect the transition to future states. Markov chains are Markov processes which operate at discrete time intervals and within a discrete state space. The computational domain Ψ of a horizontal rotating drum is decomposed into N cells based on the location, as shown in Fig. 2. The state variables are the fraction of number of particles in each cell (i.e., the number of particles in each cell divided by the total number of particles in the drum). As such, there are N states in the Markov chains model. For analysis purposes, the states are stacked to form a column vector called Markovian states, $\mathbf{s} = \left[s^1 \ s^2 \ \ldots \ s^N\right]^T$,

Table 1
Physical parameters and their values.

Parameters	Values
Drum, $D \times L$ (mm)	100×16
Particle Size, d (mm)	3
Particle density, ρ ($kg \cdot m^{-3}$)	2.5×10^3
Loading, N_p (# particles, % filling)	1000 (18%), 2000 (35%)
Drum rotational speed, Ω (rpm)	5–330
Number of cells, N	88

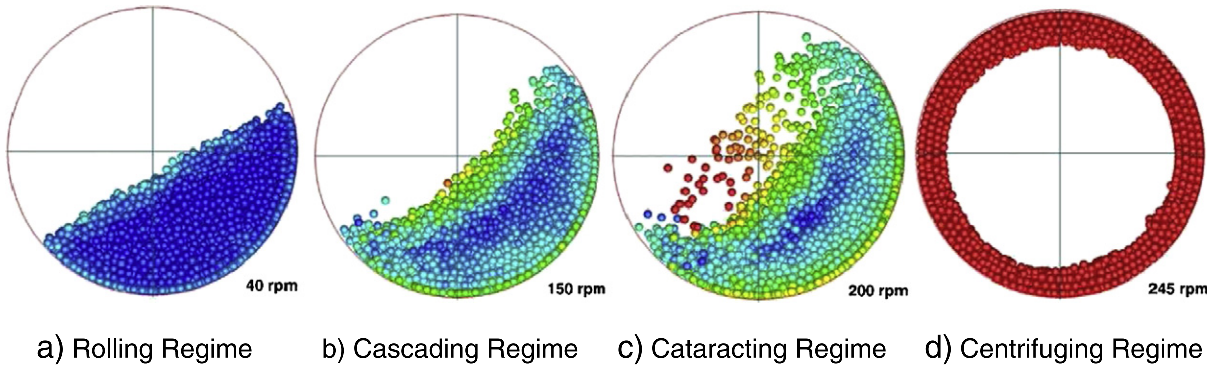

Fig. 1. Particle flow regimes for various operating conditions in a rotating drum with 2000 particles [1].

consisting of elements which have a value between 0 and 1. Superscript T refers to the transpose of a vector.

Here, we consider how the states evolve with time after each drum revolution. Denote $\mathbf{s}_k = [s_k^1\ s_k^2\ \ldots\ s_k^N]^T$ as the distribution of the number of particles in different cells after k revolutions (called k time steps in this paper), which are dependent only on the previous states of the system $\mathbf{s}_{k-1}$ mapped through a Markov chain operator [18,28,29]. Each element in Markov chain operator refers to the transition probability of particle movement from cell i to cell j, $p_{ij,k}$, after one drum revolution (from the time step of $k-1$ to k). Let $\phi_p(i,k)$ denote an indicator function which takes the value 1 if particle p is in cell i at time k, and zero otherwise. Then the elements of the Markov chain operators to be constructed in this work, denoted as $p_{ij,k}$, is obtained by using the probability of transition of a particle p from the cell i at iteration $k-1$ to cell j at iteration k, i.e.,

$$Pij, k = P\left\{\phi_p(j,k) = 1 \middle| \phi_p(i,k-1) = 1\right\}. \tag{3}$$

As in [14,17,18,30], the process is assumed to be stationary, so that the transition probability $p_{ij,k}$ is assumed to be constant for each transition time step, i.e., $p_{ij,k} = p_{ij}$.

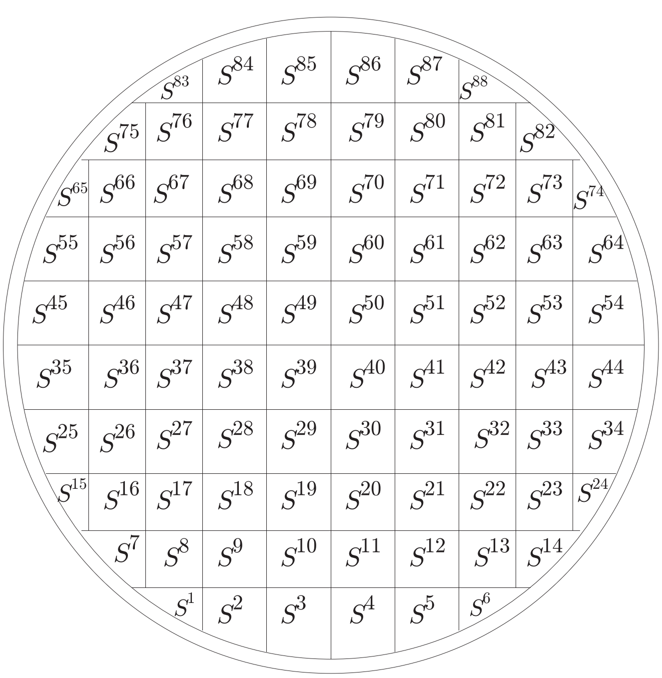

Fig. 2. The side view of horizontal rotating drum. Computational domain Ψ is divided into N number of cells, here $N = 88$.

The Markov property implies that the state of the chain can be written as

$$\mathbf{s}_{k+1} = \mathbf{P}\mathbf{s}_k, \tag{4}$$

where $\mathbf{P}$ is the Markov chain operator or the transition probability matrix. By taking successive iterates of the Markov chain operator on the initial state vector $\mathbf{s}_0$, Eq. (4) can be represented as:

$$\mathbf{s}_k = \mathbf{P}^k\mathbf{s}_0, \tag{5}$$

where $\mathbf{P}^k$ refers to k times multiplication of matrix $\mathbf{P}$. The elements of $\mathbf{P}$ are the probability of particle movement between two different states or the probability of particle staying in the same state, given as

$$\mathbf{P} = \begin{bmatrix} p_{ii} & \cdots & p_{ji} \\ \vdots & \ddots & \vdots \\ p_{ij} & \cdots & p_{jj} \end{bmatrix} \tag{6}$$

The matrix $\mathbf{P}$ has the property that the summation of the elements over each column vector is unity and the value of each element is always greater or equal to zero and less than or equal to one because these elements are probability measures of particle movement [14–17].

Remark 1. After a large number of k, $\mathbf{P}$ matrix does not change with further empowering and will approach a matrix of which each column represents the final particle distribution (referred to as the stationary distribution [19,20]). In this context, the limit of $\mathbf{P}^k$ as k becomes large provides the information on the *steady state particle distribution* and the number of iterations it requires to reach such a steady state.

3.2. Construction of the Markov chain operator

In this paper, particulate trajectory data generated from DEM simulations are used to compute each element of matrix $\mathbf{P}$. The total number of particles in cell i at time step k is first computed as

$$\varphi(i,k) = \sum_{p=1}^{N_p} \phi_p(i,k), \tag{7}$$

where N_p is the total number of particles in Ψ. Then the probability transition of particle movement from cell i to j, p_{ij}, can be obtained by

$$p_{ij} = \frac{1}{N_{LT}} \sum_{k=1}^{N_{LT}} \frac{1}{\varphi(i,k-1)} \sum_{p=1}^{N_p} \phi_p(j,k)\phi_p(i,k-1). \tag{8}$$

The information on the particle trajectories collected within a sufficient number of time steps is required to construct the $\mathbf{P}$ matrix, which describes the possibility of particle movements. Such a process is referred to as a “learning process”, as shown in Eq. (8), where the number of the time steps required is denoted as N_{LT}. The value of N_{LT} should be chosen such that the difference between $\mathbf{P}(N_{LT})$ and $\mathbf{P}(N_{LT}+1)$, quantified as

$$\epsilon(N_{LT}) = \frac{||\mathbf{P}(N_{LT}) - \mathbf{P}(N_{LT}+1)||_1}{||\mathbf{P}(N_{LT})||_1}, \tag{9}$$

is sufficiently small (as in [18]). In this work, N_{LT} is chosen to be between 30 and 40, leading to values of $\epsilon(N_{LT})$ that are close to zero. The elements of the $\mathbf{P}$ matrix, $p_{i,j}$ will converge to constants, when the time step $k \to N_{LT}$. In addition, the dynamics of the states are represented as a function of the time steps (the number of revolutions of the drum, called the sampling period) rather than the actual time so that the results with different drum rotation speeds can be compared.

In the next section, approaches to extract collective dynamical features, such as dynamic modes of the oscillatory and spatial distribution of particle movements, are introduced.

4. Analysis of Markov chain operators

This section presents two approaches, based on eigenvalues and singular values analysis, to quantitatively characterize the collective dynamical features from the matrix $\mathbf{P}$. The proposed analysis considers the dynamic modes of the state's oscillatory behavior, frequency distribution, collective dynamics index and spatial particulate distribution.

The largest eigenvalue of the matrix $\mathbf{P}$ corresponds to the steady state behavior and the second largest eigenvalue was used to study the state convergence rate, known as the mixing time [19,20]. This work extends the eigenvalue analysis to quantitatively characterize the collective features of oscillatory behavior dynamic modes of particle distributions. The readers are referred to [31,32] for matrix theory of eigenvalues. Furthermore, SVD is used in this work to extract the spatial distribution of particulate movement from the Markov chain operator. The readers are referred to [33] for fundamental theory of SVD. The reported studies of Markov chain operator based on SVD in the literature are very limited. Approaches based on SVD have been used for model reduction and identification purposes of Markov chain systems [34,35]. But none of the SVD based approaches are used to study the system collective dynamical features.

4.1. Modes of collective dynamics

For each drum rotational speed, the behavior of the contents can be quantified by dynamic modes obtained from the $\mathbf{P}$ matrix. This section introduces a method to analyze the oscillatory behavior in terms of dynamic modes using the magnitude and the normalized frequency of oscillation obtained from eigenvalues of the $\mathbf{P}$ matrix.

We start formulating the approach by presenting the following concepts. The eigendecomposition of the matrix $\mathbf{P}$ can be presented as,

$$\mathbf{P}\mathbf{v}_i = \lambda_i \mathbf{v}_i, \quad i = 1, \ldots, N, \tag{10}$$

where λ_i is the i-th eigenvalue of the $\mathbf{P}$ matrix and $\mathbf{v}_i$ is its corresponding eigenvector. So, the initial particle distribution, $\mathbf{s}_0$, can be represented as the linear combination of all eigenvectors:

$$\mathbf{s}_0 = a_1\mathbf{v}_1 + a_2\mathbf{v}_2 + \ldots + a_N\mathbf{v}_N, \tag{11}$$

where $a_i \in \mathbb{C}$ are coefficients associated to the eigenvectors v_i, $i = 1, \ldots, N$. Using Eq. (5), we have:

$$\begin{aligned}\mathbf{s}_k &= \mathbf{P}^k(a_1\mathbf{v}_1 + a_2\mathbf{v}_2 + \ldots + a_N\mathbf{v}_N) \\ &= a_1\mathbf{P}^k\mathbf{v}_1 + a_2\mathbf{P}^k\mathbf{v}_2 + \ldots + a_N\mathbf{P}^k\mathbf{v}_N,\end{aligned} \tag{12}$$

and based on Eq. (10),

$$\mathbf{s}_k = a_1(\lambda_1)^k\mathbf{v}_1 + a_2(\lambda_2)^k\mathbf{v}_2 + \ldots + a_N(\lambda_N)^k\mathbf{v}_N, \tag{13}$$

where $(\lambda_N)^k$ is the k-th power of the N-th eigenvalue of the $\mathbf{P}$ matrix. Eq. (13) describes the dynamics of the particle distribution, as the linear combination of the terms $(\lambda_i)^k\mathbf{v}_i$, $(i = 1, \ldots, N)$. As such, $(\lambda_i)^k$, as a function of time steps, represents a *dynamic mode* of the collective dynamics of the system, on the direction of $\mathbf{v}_i$.

The eigenvalues can be complex. Assuming $\lambda_i = \alpha_i + j\beta_i$, where $j = \sqrt{-1}$, we have:

$$\begin{aligned}(\lambda_i)^k = (\alpha_i + j\beta_i)^k &= (r_i)^k e^{jk\omega_i} \\ &= (r_i)^k(\cos(k\omega_i) + j\ \sin(k\omega_i)),\end{aligned} \tag{14}$$

where

$$r_i = \sqrt{\alpha_i^2 + \beta_i^2} \tag{15}$$

$$\omega_i = \tan^{-1}\left(\frac{\mathrm{Im}(\lambda_i)}{\mathrm{Re}(\lambda_i)}\right) = \tan^{-1}\left(\frac{\beta_i}{\alpha_i}\right). \tag{16}$$

The magnitude and frequency of oscillation for dynamic mode i of the system are represented by r_i and ω_i, respectively. The four-quadrant arctangent is used in Eq. (16) (e.g., the "atan2" function in MATLAB) so that ω is in the interval of $[-\pi, \pi]$. A negative real eigenvalue is associated with fastest possible frequency of π, meaning that the corresponding dynamic mode oscillates every time step, finishing a cycle in two time steps.

The ω_i in Eq. (14) is a normalized frequency of the dynamic mode i. The actual frequency, denoted as $\tilde{\omega}_i$, can be calculated as follows: the time step is $k = t/\Delta t$, where t is the actual time and Δt is sampling period, which is $\Delta t = 1/\Omega$ (Ω is the drum rotational speed, in the unit of rpm). Therefore, $\tilde{\omega}_i = \omega_i\Omega$. Denote the angular frequency of the drum rotation as $\omega_d = 2\pi\Omega$. We have

$$\tilde{\omega}_i = \omega_i\frac{\omega_d}{2\pi}. \tag{17}$$

As the maximum ω_i is π, the actual frequency of the fastest dynamic mode is $\tilde{\omega}_i = \omega_d/2$.

Substituting Eq. (14) into Eq. (13), we have the total dynamics of the particle distribution as follows:

$$\begin{aligned}\mathbf{s}_k = {} & a_1(r_1)^k(\cos(k\omega_1) + j\ \sin(k\omega_1))\mathbf{v}_1 + a_2(r_2)^k(\cos(k\omega_2) + j\ \sin(k\omega_2))\mathbf{v}_2 + \ldots \\ & + a_N(r_N)^k(\cos(k\omega_N) + j\ \sin(k\omega_N))\mathbf{v}_N\end{aligned} \tag{18}$$

$$= a_1(r_1)^k(\cos(k\omega_1))\mathbf{v}_1 + a_2(r_2)^k(\cos(k\omega_2))\mathbf{v}_2 + \ldots + a_N(r_N)^k(\cos(k\omega_N))\mathbf{v}_N. \tag{19}$$

Complex eigenvalues always appear in pairs, which are complex conjugates. That is, if one eigenvalue is $\alpha + j\beta$, then there is another eigenvalue of $\alpha - j\beta$. The corresponding eigenvectors are complex conjugates of each other. For example, if λ_1 and λ_2 are a pair of complex eigenvalues, then their corresponding eigenvectors $\mathbf{v}_1$ and $\mathbf{v}_2$ are complex conjugates. In this case, the initial particle distribution on the directions of $\mathbf{v}_1$ and $\mathbf{v}_2$ will have the same coefficients $a_1 = a_2$ as in Eq. (11). Therefore, the imaginary parts in Eq. (18) will cancel out, leading to Eq. (19).

Clearly the eigenvalues represent the features of the collective dynamics. By definition, all eigenvalues of the Markov chain operator, $\mathbf{P}$, have the magnitude $r_i \leq 1$ (i.e., within or on the unit circle) [19,20]. For the dynamic mode associated with λ_i: the modulus r_i tells its magnitude (or magnitude of oscillation if $\omega_i \neq 0$), e.g., how fast the dynamics decay when $r_i < 1$ (if $r_i = 1$, then the dynamic mode persists);

ω_i indicates its normalized frequency of oscillation. In [19,20], the second largest modulus of the eigenvalue (which is less than 1) is used to indicate the "mixing time". Here, we use both the modulus and oscillatory frequencies of all dynamic modes to describe the collective dynamical behavior of particle distribution.

The oscillatory behaviors in the collective dynamics are linked to the locations of the eigenvalues on a complex plane. For illustrative purposes, the complex eigenvalues for 1000 and 2000 particulate systems at various operating conditions are presented in Figs. 3 and 4 respectively. In these figures, the horizontal and vertical axes are the real part and imaginary part of the eigenvalues, respectively. Therefore, r_i is the distance from the origin and ω_i is the angle between the horizontal axis and the line from the location of the eigenvalue to the origin. It can be observed that the modulus of the eigenvalues with nonzero frequencies increases with the drum rotational speed.

As shown in Figs. 3 and 4 at low drum rotational speeds, the eigenvalues are concentrated close to the origin. As the drum rotational speed increases, some of the eigenvalues start to move away from the origin. This shows an increase in the magnitude of oscillations represented by the distance from the origin to the eigenvalues values r_i. Finally, all eigenvalues are located either at the origin or close to the unit circle when the centrifugal force dominates the system.

Most eigenvalues are located on the positive real axis very close to 1 for a 1000 particle system operated at 290 rpm and above, indicating that particle distribution in each cell remains relatively constant due to dominant centrifugal force. This is consistent with results reported in [1,21] which are summarized in Table 2. Similar results are observed in 2000 particle systems. This shows that the complex eigenvalues of **P** can be linked to the flow regimes.

The frequency spectrum of the dynamic modes are shown in Figs. 5 and 6 for 1000 and 2000 particulate systems respectively. It can be seen that the dynamic modes with largest magnitude shifted to high frequencies with increase in drum rotational speed. The shift is affected by the particle loading. It is shown that the dominant normalized frequency for 1000 particles is around 2.8 rad while for 2000 particles is around 2.5 rad. The system reached the centrifuging regime at 290 rpm for 1000 particles and 230 rpm for 2000 particles (as reported in [22,24]). In the centrifuging regime, all dynamic modes have zero frequency, indicating that there is no change in particle distribution with time.

A *collective dynamics index* (CDI) which describes the overall oscillatory behaviors is constructed based on the magnitudes, $\mathbf{r} = [r_1\ r_2\ \dots\ r_N]^T$, and normalized frequencies, $\omega = [\omega_1\ \omega_2\ \dots\ \omega_N]^T$, of all dynamical modes. Denote

$$\hat{\omega} = [|\omega_1|\ \ |\omega_2|\ \ \dots\ \ |\omega_N|]^T. \tag{20}$$

The CDI is quantified as follows:

$$\mathrm{CDI} = \langle \mathbf{r}, \hat{\omega} \rangle = \mathbf{r}^T \hat{\omega} \tag{21}$$

$$= \sum_{i=1}^{N} r_i |\omega_i|. \tag{22}$$

Fig. 7a and b show the values of CDI against drum rotational speed for 1000 and 2000 particles, respectively. It can be observed that the CDI increases with increasing rotational drum speed. A sudden drop close to zero is observed and the index remains around zero with further increase in the drum rotational speed. This is to be expected, since the oscillatory behavior in particle distribution increases with increasing drum rotational speed, and as it reaches a certain limit, the centrifugal force dominates the system and as a result the particle distribution in each cell remains relatively constant. It is interesting to

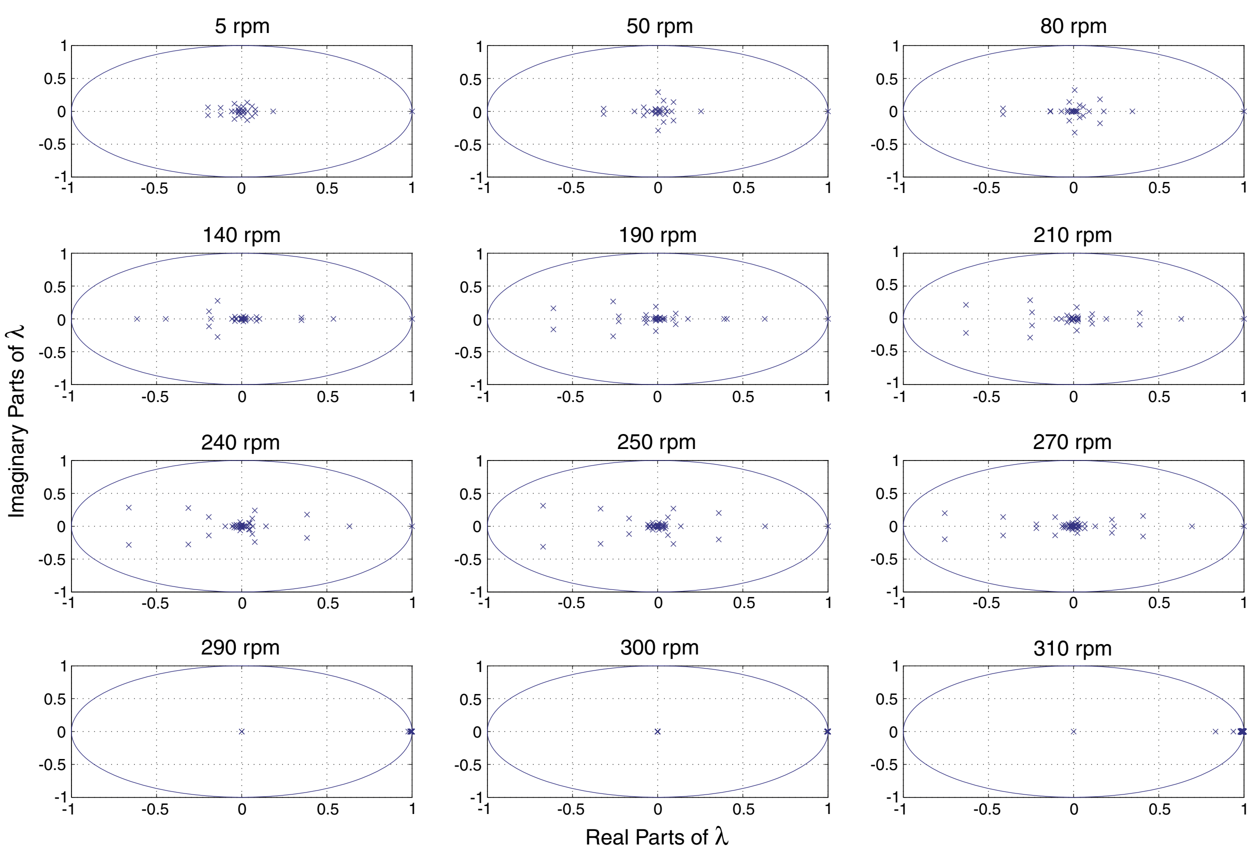

Fig. 3. Eigenvalue distribution of a 1000 particulate system in the complex plane for different operating conditions.

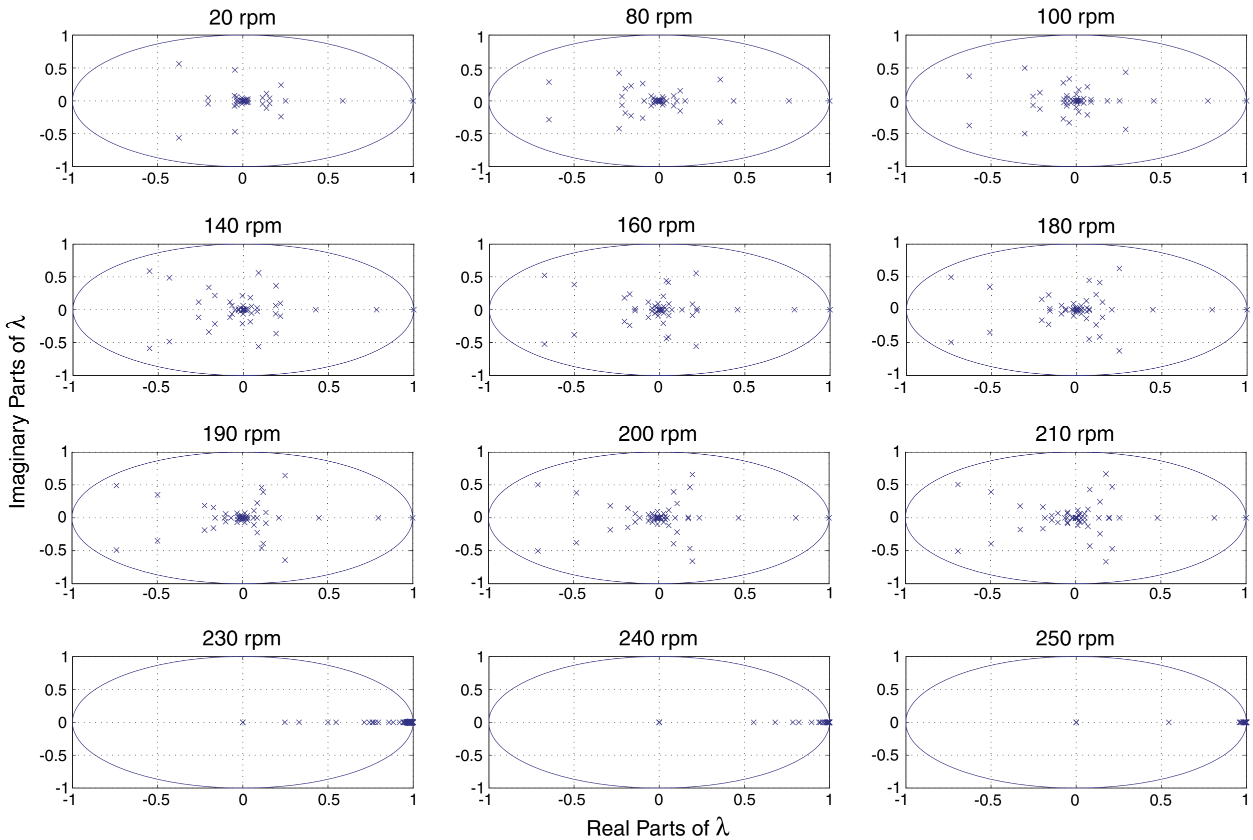

Fig. 4. Eigenvalue distribution of a 2000 particulate system in the complex plane for different operating conditions.

see that the CDIs are clearly related to the flow regimes reported in [22,24], summarized in Table 2.

4.2. Spatial distribution of particle movements

An approach to study the spatial distribution of particulate movements is presented in this section. It is used to analyze the particulate movement and determine the locations where major particle movements occur. This approach is based on the analysis of singular values.

The change of particle distribution between two time steps, from k to $k+1$ can be described as follows:

$$\begin{aligned}\Delta\mathbf{s}_{k+1,k} &= \mathbf{s}_{k+1}-\mathbf{s}_k = \mathbf{P}\mathbf{s}_k-\mathbf{s}_k\\ &= (\mathbf{P}-\mathbf{I})\mathbf{s}_k\end{aligned} \tag{23}$$

where $\mathbf{I}$ is an identity matrix. Matrix $(\mathbf{P}-\mathbf{I})$ can be understood as a mapping from the input $\mathbf{s}_k$ to the output $\Delta\mathbf{s}_{k+1,k}$. It can be decomposed using SVD [33] as follows:

$$\mathbf{P}-\mathbf{I} = \mathbf{U}\Sigma\mathbf{X}^T \tag{24}$$

Table 2
Flow regime classification based on visual observation of DEM simulations [22,24].

1000 particles		2000 particles	
Flow regime	Drum speed (rpm)	Flow regime	Drum speed (rpm)
Rolling	$0\le\Omega<140$	Rolling	$0\le\Omega<120$
Cascading	$140\le\Omega<240$	Cascading	$120\le\Omega<180$
Cataracting	$240\le\Omega<290$	Cataracting	$180\le\Omega<230$
Centrifuging	$290\le\Omega$	Centrifuging	$230\le\Omega$

where $U\in\mathbb{R}^{N\times N}$ and $X^T\in\mathbb{R}^{N\times N}$ are unitary matrices and $\Sigma\in\mathbb{R}^{N\times N}$ is a diagonal matrix of ordered singular values, where $\sigma_1>\sigma_2>\ldots>\sigma_N\ge 0$. Therefore Eq. (23) can be written as

$$\begin{aligned}\Delta\mathbf{s}_{k+1,k} &= \mathbf{U}\Sigma\mathbf{X}^T\mathbf{s}_k\\ &= [\mathbf{u}_1 \cdots \mathbf{u}_N]\begin{bmatrix}\sigma_1 & & 0\\ & \ddots & \\ 0 & & \sigma_N\end{bmatrix}[\mathbf{x}_1 \cdots \mathbf{x}_N]^T\mathbf{s}_k\\ &= \left[\sigma_1\ \mathbf{u}_1\ \mathbf{x}_1^T\ + \cdots + \sigma_N\ \mathbf{u}_N\ \mathbf{x}_N^T\right]\mathbf{s}_k.\end{aligned} \tag{25}$$

where $\mathbf{u}_i = [\mathrm{u}_{i1}\ \mathrm{u}_{i2}\ \ldots\ \mathrm{u}_{iN}]^T$ and $\mathbf{x}_i = [\mathrm{x}_{i1}\ \mathrm{x}_{i2}\ \ldots\ \mathrm{x}_{iN}]^T$ are left and right singular vectors, respectively, corresponding to the ith singular value, σ_i. In this work, $N=88$ corresponding to the number of cells in the computational domain Ψ. Because $\mathbf{U}$ and $\mathbf{X}$ are unitary matrices, we have $\mathbf{U}^T\mathbf{U}=\mathbf{X}^T\mathbf{X}=I$. The set of vectors $\mathbf{u}_i$, $i=1, \ldots, N$ form an orthonormal basis for the vector space of the output $\Delta\mathbf{s}_{k+1,k}$ and the set of vectors $\mathbf{x}_i$, $i=1, \ldots, N$, forms an orthonormal basis for the vector space of the input $\mathbf{s}_k$. Any $\mathbf{s}_k$, can be represented as a linear combination of the $\mathbf{x}$ vectors,

$$\begin{aligned}\mathbf{s}_k &= \alpha_1\mathbf{x}_1 + \cdots + \alpha_N\mathbf{x}_N\\ &= \mathbf{X}\alpha,\end{aligned} \tag{26}$$

where $\alpha = [\alpha_1, \ldots, \alpha_N]^T$ is the vector of the coefficients. The net changes in particle distribution after one time step shown in Eq. (25) can be represented as,

$$\begin{aligned}\Delta\mathbf{s}_{k+1,k} &= \mathbf{U}\Sigma\mathbf{X}^T\mathbf{X}\alpha\\ &= \mathbf{U}\Sigma\alpha\\ &= \alpha_1\sigma_1\ \mathbf{u}_1 + \cdots + \alpha_N\sigma_N\ \mathbf{u}_N.\end{aligned} \tag{27}$$

Therefore, $\Delta\mathbf{s}_{k+1,k}$ is the linear combination of the left singular vectors $\mathbf{u}_i$ with a coefficient of $\alpha_i\sigma_i$, for $i=1, \ldots, N$.

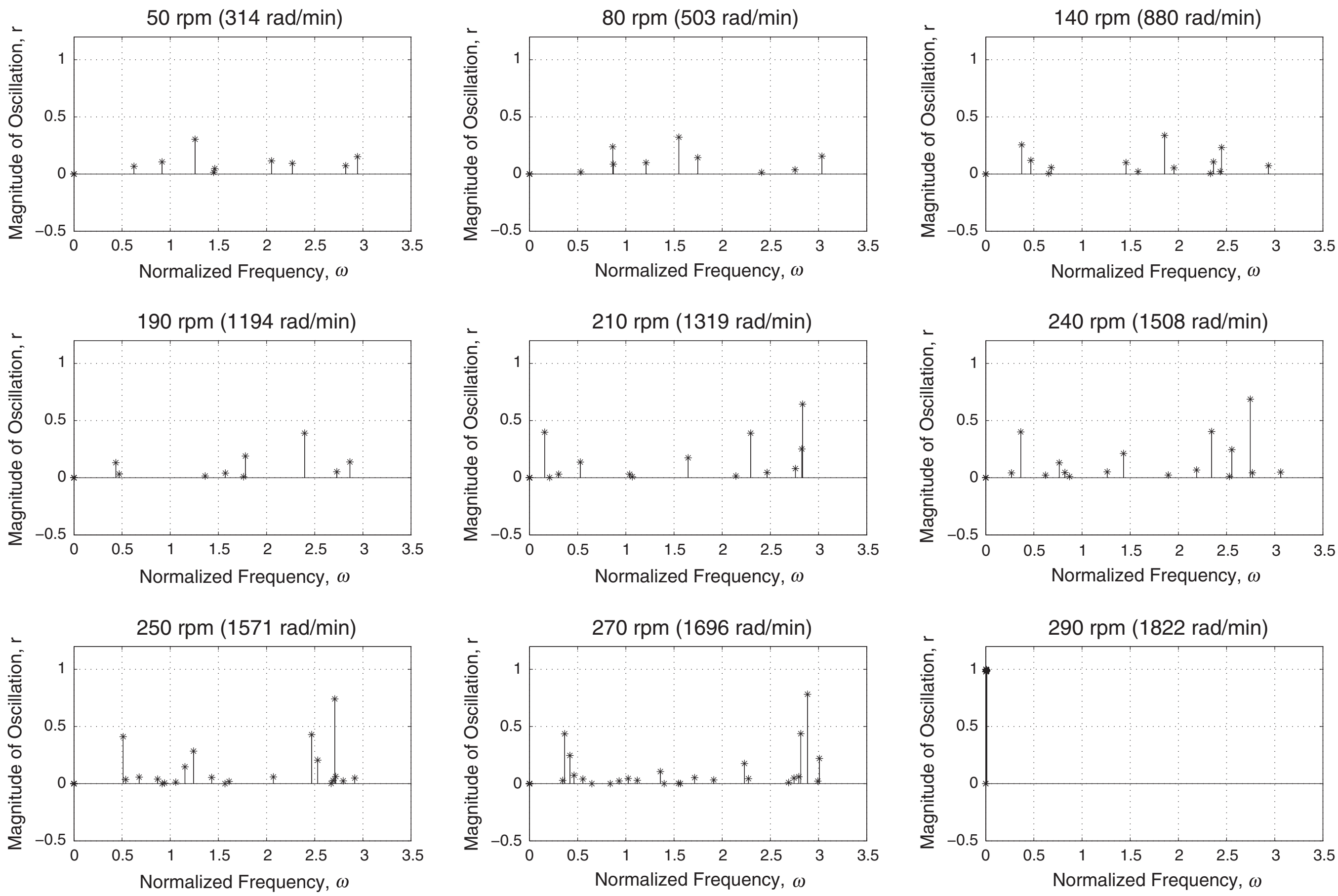

Fig. 5. Frequency distribution of dynamic modes for 1000 particles.

Consider the maximum net change in spatial particle distribution. The magnitude of total particle distribution and its net change can be quantified using the vector 2-norm, given as:

$$\|\Delta \mathbf{s}^{max}_{k+1,k}\|_2 = \|\alpha_1 \sigma_1 \ \mathbf{u}_1\|_2, \quad \|\mathbf{s}^{max}_{k}\|_2 = \|\alpha_1 \mathbf{x}_1\|_2. \tag{28}$$

Then we have

$$\frac{\|\Delta \mathbf{s}^{max}_{k+1,k}\|_2}{\|\mathbf{s}^{max}_{k}\|_2} = \frac{\|\alpha_1 \sigma_1 \ \mathbf{u}_1\|_2}{\|\alpha_1 \mathbf{x}_1\|_2} = \sigma_1, \tag{29}$$

where $\|\mathbf{x}_1\|_2 = \|\mathbf{u}_1\|_2 = 1$ and σ_1 indicates the maximum net change in particle distribution after one time step in relation to the previous step particle distribution (the "gain" of the mapping), in terms of their 2-norms. The maximum change in the particle distribution occurs on the direction given by $\mathbf{u}_1$ (with a scalar multiplier of $\alpha_1\sigma_1$) which shows the "pattern" of the changes of particle fractions in each cell related to the largest net change of particle distribution. Assigning each element of $\mathbf{u}_1$ based on the cell geometry shown in Fig. 2, the locations of the net changes in particle fractions (movement) can be determined.

A system consisting 2000 particles is used to illustrate this approach, as shown in Fig. 8a–d. It can be observed that there are positive and negative elements corresponding to the percentage of particles moving in and out of the cells, respectively. The cells with high positive coefficients have a high number of particles moving in while the cells with high negative coefficients have a high number of particles moving out. The spatial patterns of net particle movement are very close to the reported flow regime pattern in [22,24], and clearly separates the different flow regimes. It is interesting to note that the left singular vector coefficients for 230 rpm, shown in Fig. 8d, are very close to zero. This is consistent with the observation of centrifuging flow regime where the particle distribution is constant.

5. Discussion

The study of collective dynamical features of particulate system in a horizontal rotating drum with 1000 and 2000 particles is presented. The transition probability matrix (i.e., Markov chain operator) is constructed based on data generated from DEM simulations. The physical time step of DEM simulation influences the accuracy of the DEM simulation results (e.g., particle trajectories). However, at least in theory, the time step for DEM simulation should not affect the transition matrix (i.e., Markov chains operator) if such simulation is performed with sufficient accuracy. The Markov chain operator is constructed based on the information of particle locations at "sampling time" points. In this paper, one sample is taken in every single drum revolution. The time step used in our DEM simulation is identical to the one used in [1,21] which has been experimentally validated to produce particle trajectories with high accuracy. Therefore, the Markov chain operator generated in our work has high accuracy of the particle movement probability.

Two approaches based on eigenvalues and singular value decomposition are used to quantitatively study and characterized the features of collective dynamics of the particulate system. The eigenvalue approach is used to quantitatively study various dynamic modes corresponding to oscillatory particulate behaviors, including eigenvalue analysis in complex plane, frequency distribution analysis, and collective dynamics index of state oscillatory behavior. The singular value decomposition approach is used to quantitatively study the change in spatial distribution of particulate movement. Numerical and quantitative results from these approaches to the analysis of particulate systems were compared

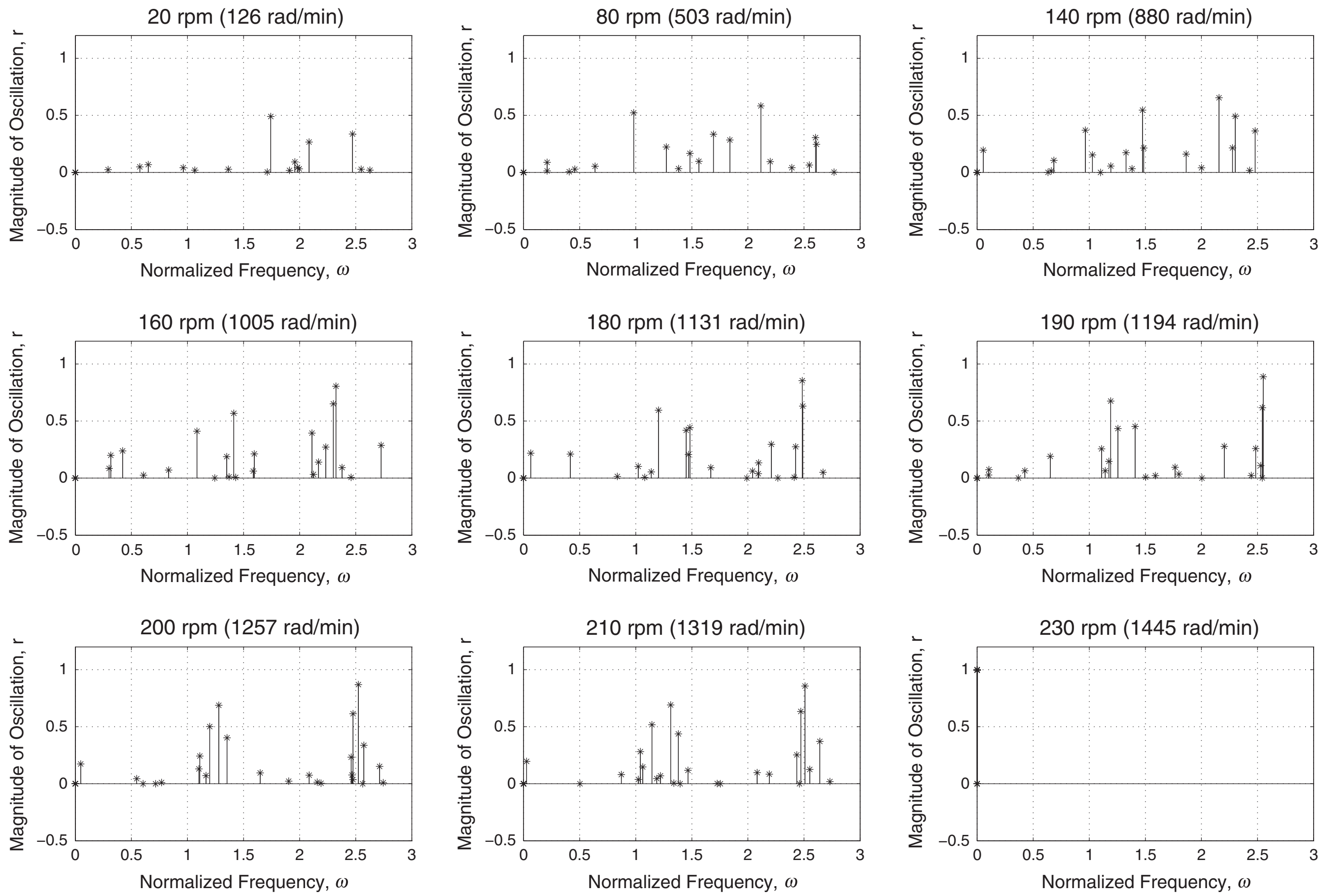

Fig. 6. Frequency distribution of dynamic modes for 2000 particles.

and proved to be consistent with qualitative flow regime results previously presented in the literature.

The qualitative approaches based on visual observation are commonly used in particulate studies to characterize the overall particle dynamics. However these techniques give coarse estimations and lack predictive capacity on collective dynamics, since they are mainly based on particle loading and drum rotational speed. This renders collective dynamics based on qualitative analysis impossible to use, for example, for process control. On the other hand, the quantitative study proposed in this work is able to provide a quantitative measure for the collective dynamics of particulate systems which opens a pathway towards process control. However, the limitation of the proposed quantitative approach is the construction of the Markov chain operator for large industrial scale systems where the DEM simulation is computationally expensive.

6. Conclusion

This paper proposed a quantitative approach to determine the collective dynamical features of particulate systems in a horizontal rotating drum modeled by Markov chains. DEM simulations are used to construct Markov chain operators. A measure based on eigenvalues

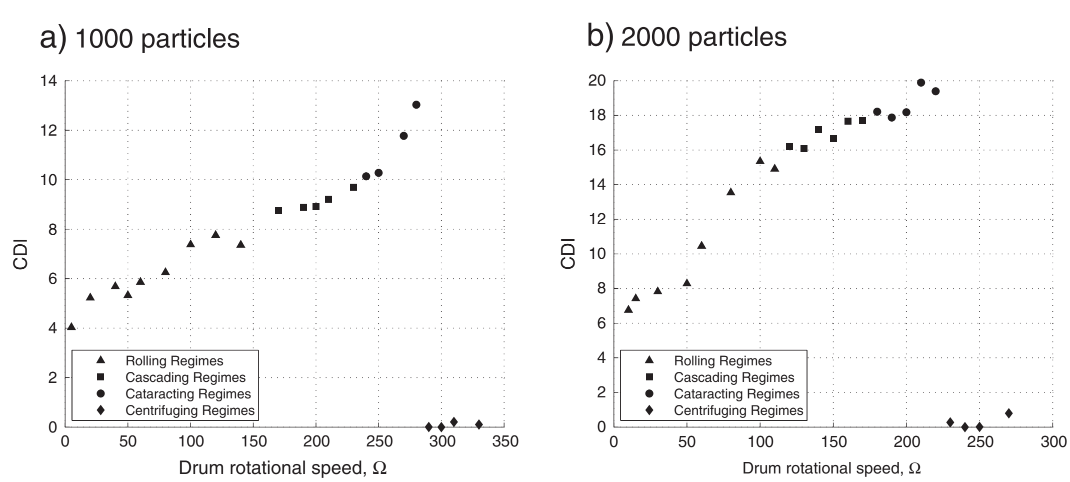

Fig. 7. Collective dynamics index (CDI) plots against drum rotational speed.

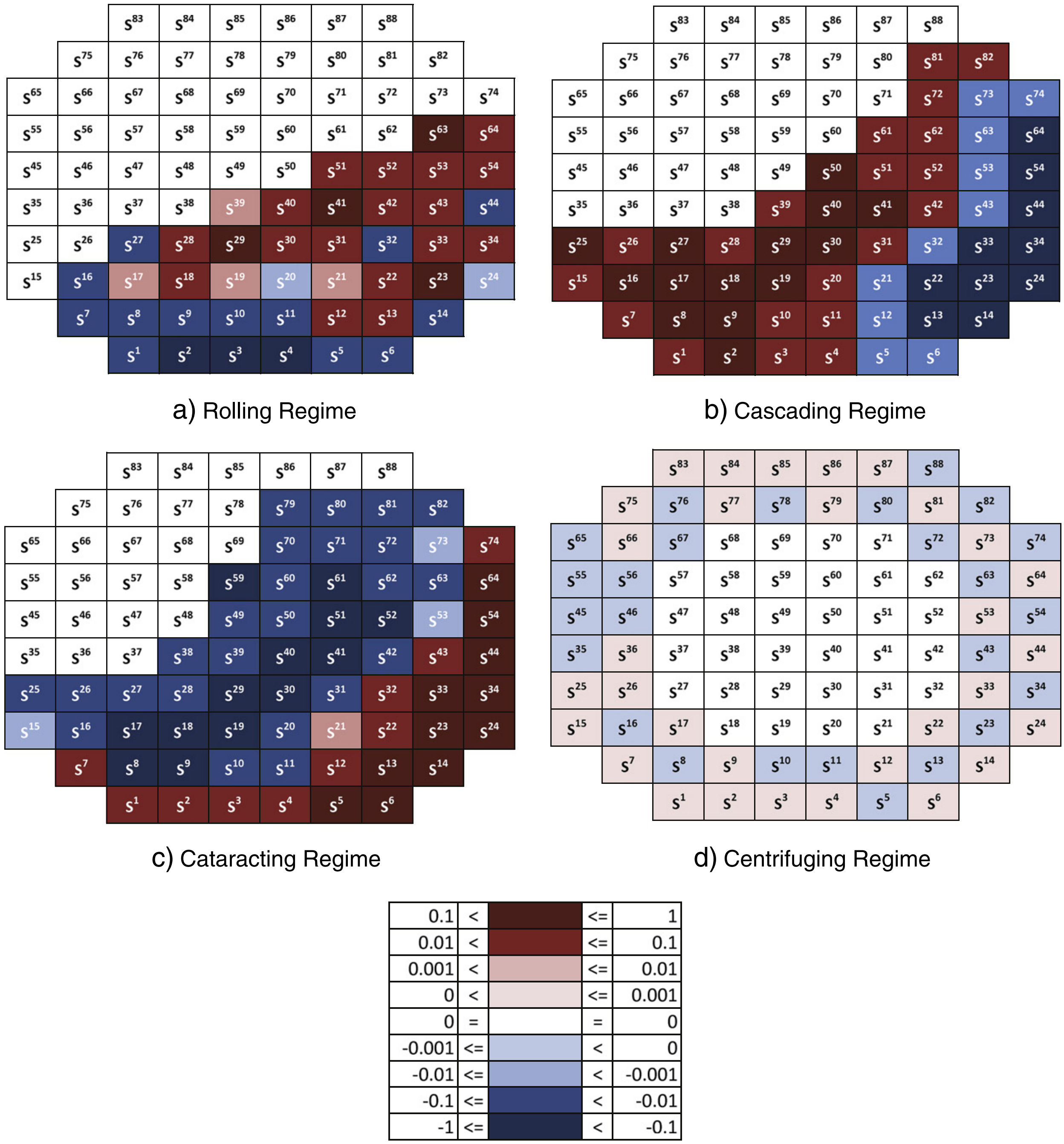

Fig. 8. Dominant spatial distribution for a 2000 particle system.

analysis of Markov chains is developed to characterize the dynamic modes of particulate systems. Additionally, a measure based on singular values is developed to characterize the spatial distribution of particle movement in particulate systems. Qualitative analyses of flow regimes from the literature are used to validate the quantitative results obtained in this paper. Current research focuses on extending the proposed approach to dynamical modeling and control of the collective dynamics of particulate systems. In particular, the problem of controlling particle distribution and flow regimes by manipulating the drum rotating speed is currently being investigated.

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